

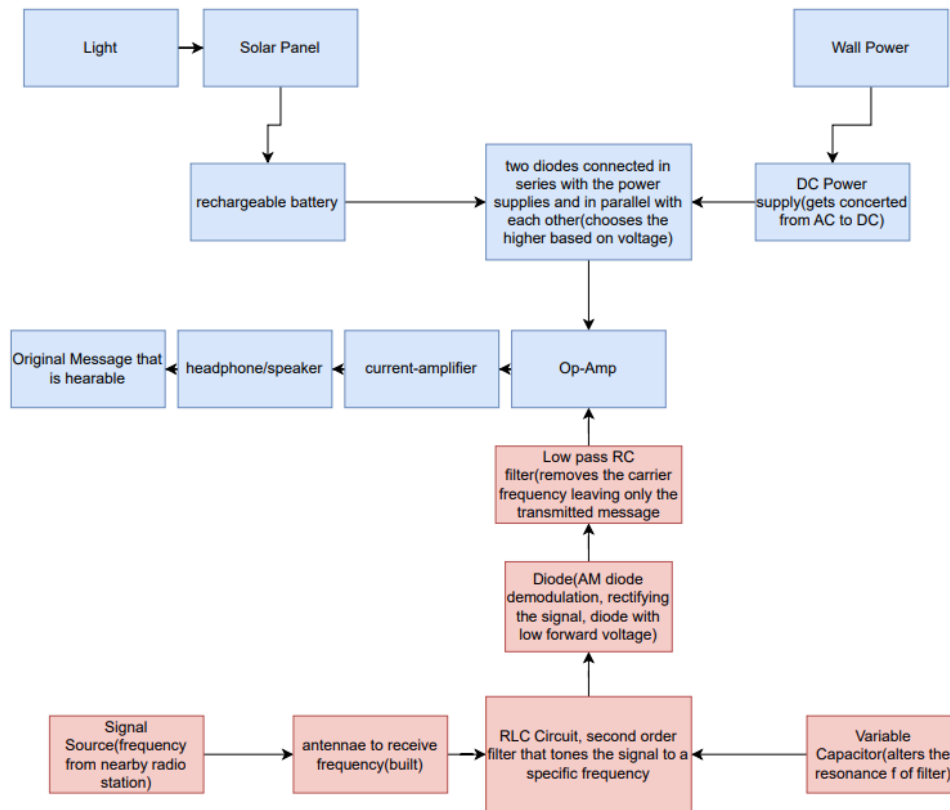
OMEGA LAB PROJECT MANUAL

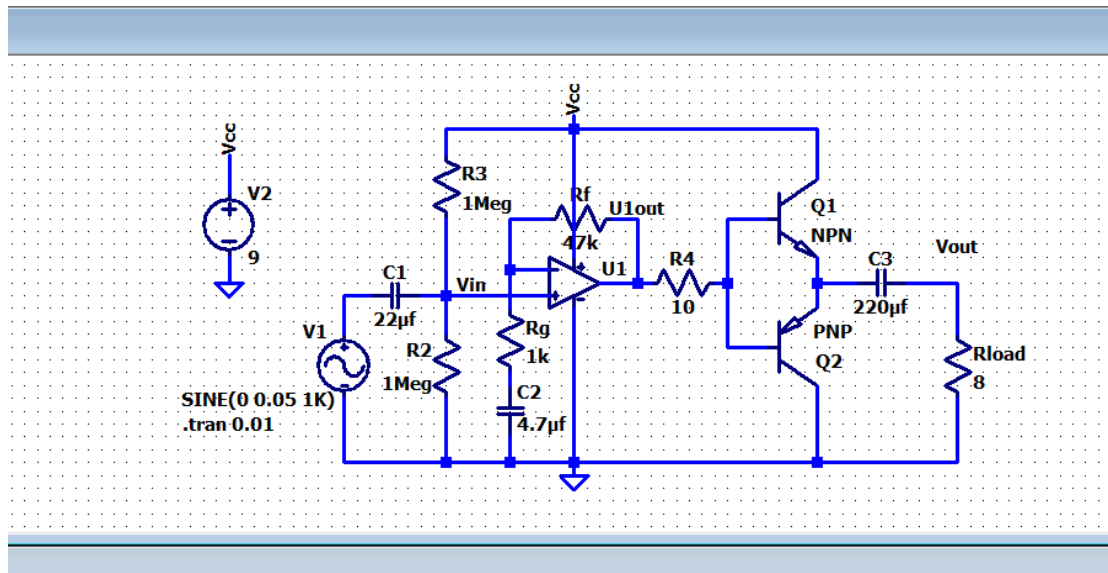
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Description

Complete schematic and block diagram

The blue represents milestone 1 and the red milestone 3





High-level description of the operation and intended application

We are trying to create a solar powered radio that can be powered via a solar panel charging a battery when other sources of voltages such as wall power are not available. The circuit will choose which of the voltage is greater between a voltage source and the solar powered panel through a diode circuit that will block the smaller voltage. This output will power an operational amplifier for use later in the circuit. We will then filter a signal received by an antenna through an RLC circuit that will be tuned to specific frequencies with a variable capacitor. The signal will then go to an op amp for voltage amplification, and transistors for current amplification. This output will then be audible through an 8 ohm speaker.

Description of related pre-existing applications

During the omega lab we found that an op amp voltage amplifier produces the correct voltage to power the speaker, however, it does not produce enough current to power the speaker, leading our group to research current amplifiers using transistors to have enough current to drive our small impedance speaker. In our design, we used a class B amplifier that is highly efficient and uses a pair of NPN and PNP transistors. Each transistor is only on during half of the input signal cycle, saving on power and heat generation. Another important part of our build was how to pick the voltage that is greater between a solar powered battery and a power supply and use it to power the signal amplification. We originally thought about using an op-amp comparator to determine which is greater but quickly gave up on that idea because we couldn't find a good way to actually switch between the two power supplies, only recognize which one was greater. Instead we implemented diodes to choose which voltage is greater as diodes possess the property of being either forward bias or reverse bias effectively either allowing current to flow or blocking current from flowing.

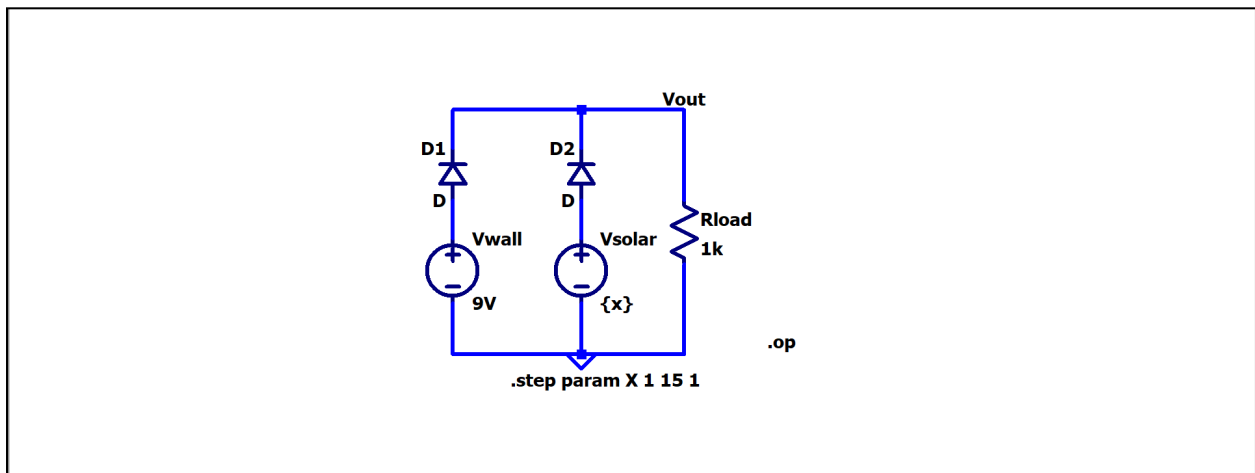
Operation and Design

Input Block: Solar Panel

A solar Panel that we got from a friend who is also working on omega labs. The solar panel is not very efficient and does not seem to be able to pick up any amount of volts when we tested it out by leaving it under the sun. Another fact that negatively impacts this building block is the fact that the solar panel at maximum can only produce a meager 3 volts, which is far from enough to power the speaker/headphone and is not really able to be compared with the wall power. Instead of using the panel in our circuit, we instead choose to simulate its effects through another source voltage.

MS1 Building Block: Diodes

Schematic



The solar panel is visualized in LTspice as a voltage source of X volts and Vwall is the other source of voltage in our circuit that is compared to the solar panel to see which is greater, we use diodes to determine which voltage gets to pass on.

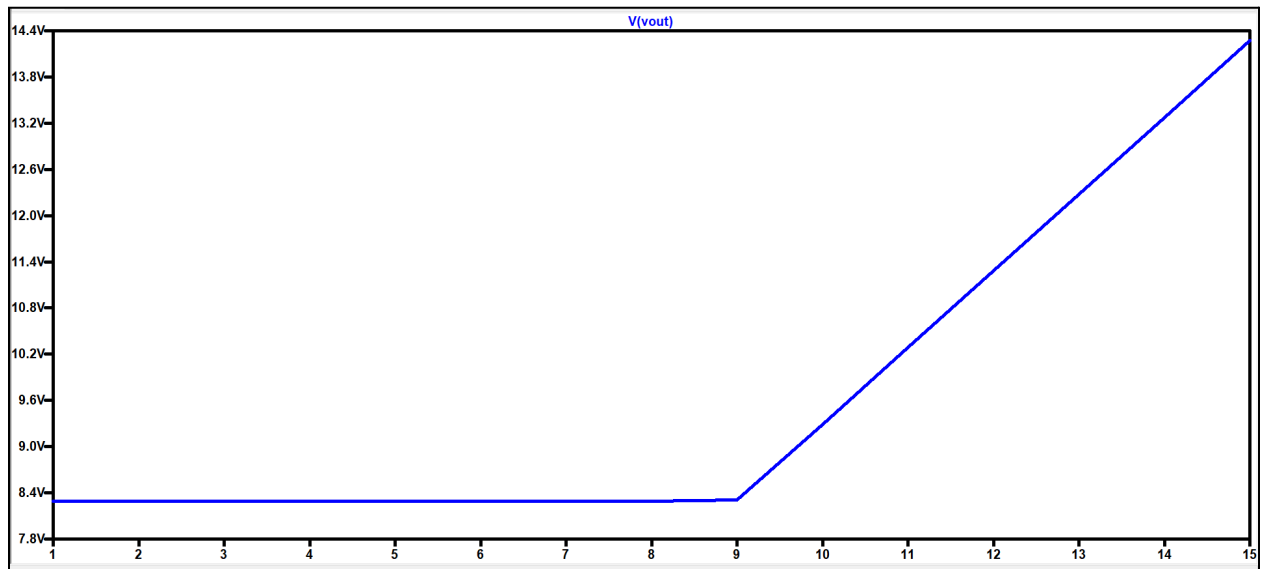
Design Equation

1. When Vwall is greater than Vsolar, the voltage at anode of D1 is greater than the cathode at the node that connects D1 and D2, thus D1 becomes forward bias allowing current to flow from Vwall to the Vout. For Vsolar in this case, since it is less than Vwall, which means that the voltage at anode of D2 is less than the cathode resulting in D2 being a reverse biased state that blocks the current from flowing.
2. When Vwall is less than Vsolar, the voltage at anode of D2 is greater than the cathode at the node that connects D1 and D2, thus D2 becomes forward bias allowing current to

flow from V_{wall} to V_{out} , and $D1$ will become a reverse biased diode that blocks current from flowing.

3. When V_{wall} is equal to V_{solar} , in this case both of the diodes are going to be forward biased and it would allow both currents to flow into the cathode resulting in a negligible difference between V_{out} and V_{wall} or V_{solar} .

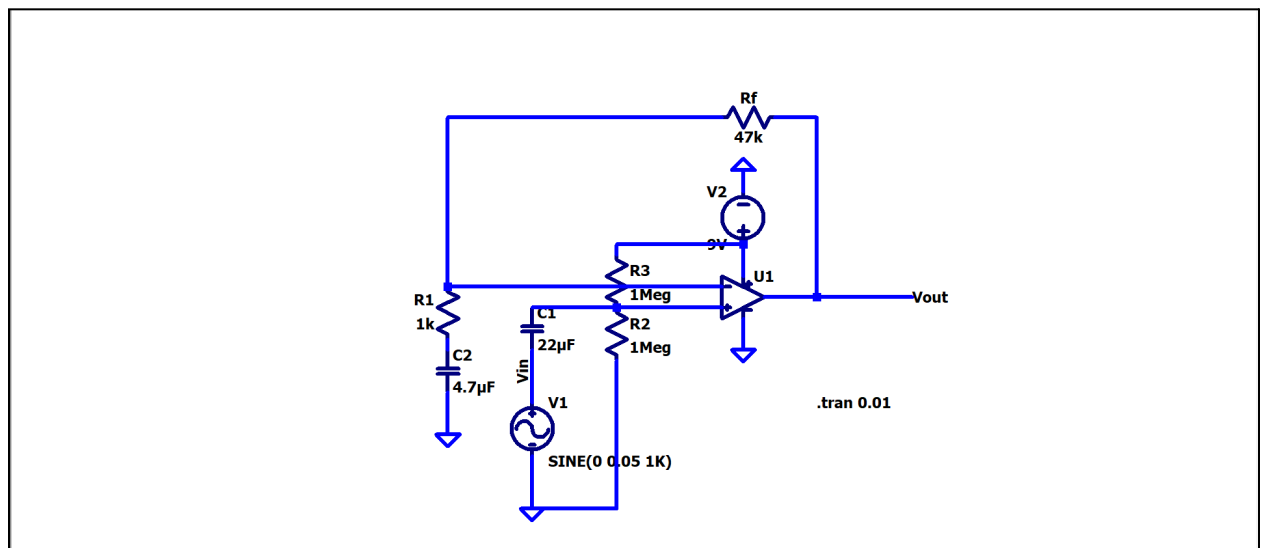
Input/Output Plot



The x-axis is the variable $\{X\}$ of the solar panel, as X becomes greater than 9 volts, the out voltage starts to grow.

MS1 Building Block 1: Non-Inverting Op Amp

Schematic



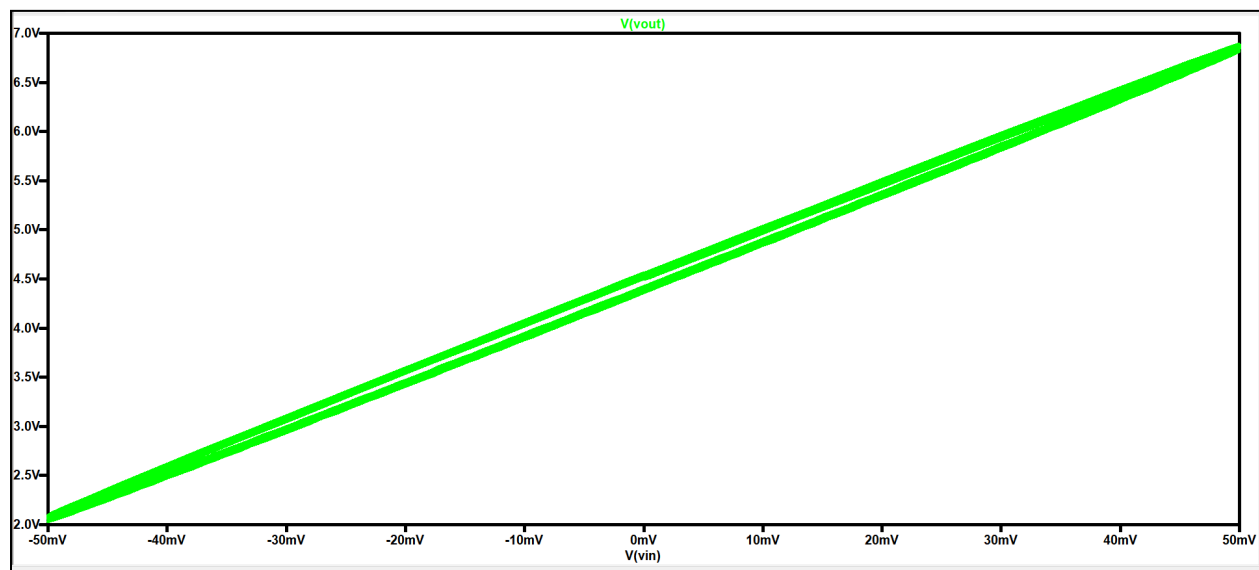
Design Equation

$$V_{out} = V_{in} \left(1 + \frac{R_f}{R_1}\right)$$

Discussion of Component Choices

We choose a resistor of 47k ohms and a resistor of 1k ohms to obtain the correct amount of gain to amplify the inputted signal to a point where it can be picked up by the 8 ohm speaker as output. We added the voltage divider to bias the input signal to 4.5 volts. $4.5V = 9V \frac{1}{2}$. This biasing was done to get the signal to be at half of the supply rails which would stop the signal from clipping during amplification. The capacitors are placed in the circuit because we need to use AC coupling to remove any remnants of DC components so that no DC can interfere and affect the amplification process, and after testing determined the best capacitance values for the two capacitors to be around 22uF and 4.7uF.

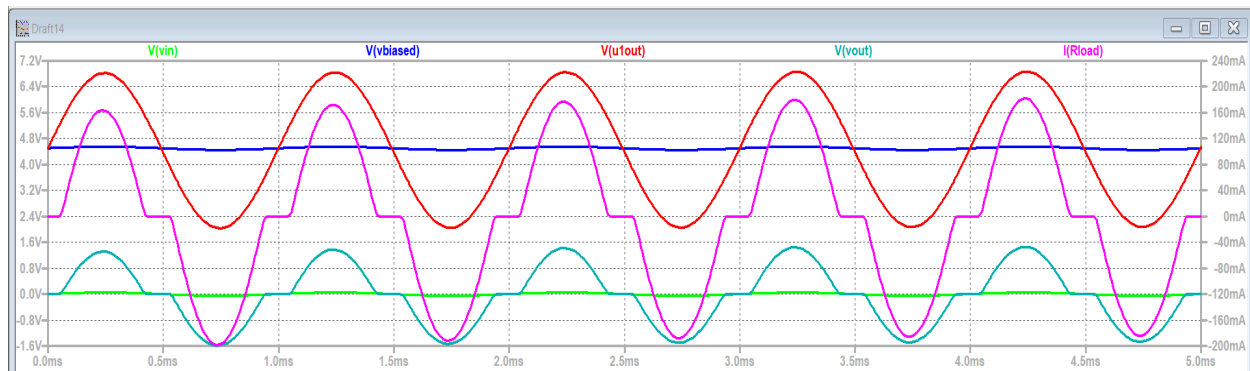
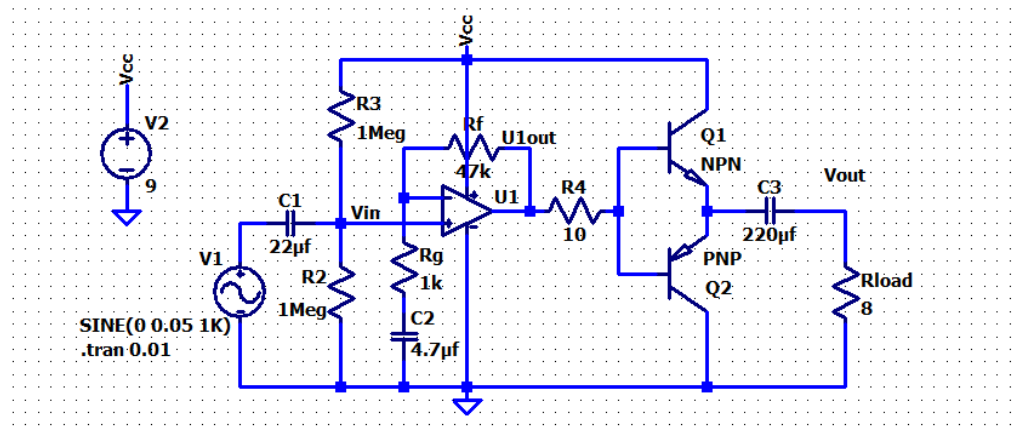
Input/Output Plot



Integration and Optimization

We have the power source via either light to solar panel or wall power that gets decided by the diode circuit. The small input signal's voltage (which will eventually come from the antenna) gets amplified by the non-inverting op-amp and the current supplied by the op-amp gets amplified by the class b amplifier. This amplified signal is then audible through the 8 ohm speaker.

Our equations compare the input sources V_{wall} and V_{solar} and the greater value will connect to the cathode that's in series, leading to the forward bias which leads the current to its next state depending on the greater value. In the case that V_{wall} is equal to V_{solar} , both currents flow to their respective cathodes. The R_{load} helped with the issue of excessive current flow throughout the circuit.



Our final inputs and outputs are graphed above. The initial signal is in green, the 4.5V biased version of this signal is in blue, the amplified signal is in red, the voltage across the speaker is in cyan and the current through the speaker is in magenta.

Operating Conditions

Describe limitations of the circuit including situations where it does not work

The circuit we built had a complex wiring design that led to an issue with transmitting a signal through the circuit. Our initial design utilized long wires that may have caused interference and degraded the overall circuit functionality. After we shortened the wires, the signal was transmitted through more efficiently and clearly. The issue taught us the importance of detailed circuit design considerations including wire length, placement, and organization to ensure the optimal performance and functionality of our circuit.

Shadows present a challenge for anything solar powered. Even in situations where the panel seems to be exposed to sunlight, shadows nearby such as buildings, trees, or passing clouds can obstruct sunlight and diminish the panel's power output, leading to inconsistent energy generated. Our output may receive interruptions or operate at a reduced capacity, particularly throughout the day where there are varying levels of shade.

Describe any trade offs present in the design

Our circuit uses a dual power source. While solar power is more renewable and environmentally friendly, it may not deliver consistent or sufficient energy. The reliance on natural elements can lead to interruptions in usage or more expensive solar panels to maintain efficiency. Using our other source of energy, wall power is more reliable and consistent, however the convenience comes at the cost of sustainability and environmental impact since the wall relies on non-renewable fossil fuels to generate the energy.

Describe how the circuit could be improved

We can improve our circuit by getting rid of the class B amplifier and implementing a class AB amplifier by adding two diodes onto the two transistors so that less crossover distortion occurs thus improving the accuracy of our output signal.

References

[1] Electronics Tutorials, "Class B Amplifier - Class-B Transistor Amplifier Tutorial," *Basic Electronics Tutorials*, Jul. 25, 2013. https://www.electronics-tutorials.ws/amplifier/amp_6.html

Appendix

If you have large images or lengthy calculations, place them here.